SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA04 **COURSE CODE:** Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments.*

QUESTIONS: 1 **Total Marks:100**

Annexure A

Technical Presentation

Code of Conduct:

I Sathikaa K 192524449 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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Annexure A

Technical Presentation [Any one]

Question 1

Suppose an Operating System is used in a banking system handling thousands of transactions simultaneously.
Prepare a technical presentation explaining:
Types of OS suitable for this scenario
OS structure used
Role of system calls in transaction processing **(BL3)**

Question 2

Consider a multi-user Linux-based server environment supporting file sharing and device access.
Prepare a technical presentation covering:
OS functions involved in resource management
File and device system calls
Interaction between user processes and kernel **(BL3)**

Question 3

Assume a real-time embedded system used in traffic signal control.
Prepare a technical presentation explaining:
Type of OS used and justification
OS structure and components
System calls used for process control and timing **(BL3)**

Question 4

A company uses a distributed computing system across multiple locations.
Prepare a technical presentation discussing:
Type of OS and its characteristics
Architecture of distributed OS
System calls used for communication and coordination **(BL3)**

Question 5

Consider an OS managing process scheduling, memory allocation, and I/O operations in a modern computer.
Prepare a technical presentation including:
Key OS functions and their importance
Interaction between OS components
Examples of system calls for each function **(BL3)**

RUBRICS FOR CALCULATION

Technical Presentation					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness



SIMATS ENGINEERING



SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

REAL-TIME OPERATING SYSTEM FOR TRAFFIC SIGNAL CONTROL: ARCHITECTURE, PROCESS MANAGEMENT AND TIMING SYSTEM CALLS

CSA0412-OPERATING SYSTEMS

GUIDED BY: DR.RIASUDEEN

PRESENTED BY: SRITHIKA.K (192524449)

INTRODUCTION

- Traffic signal systems regulate vehicle and pedestrian movement at road intersections.
- Embedded controllers operate red, yellow, and green traffic lights.
- A Real-Time Operating System (RTOS) executes critical tasks within defined time limits.
- It manages task scheduling, process control, interrupts, and timing operations.
- Sensor inputs help the controller respond to changing traffic conditions.
- RTOS supports predictable responses and reliable traffic signal operation.

OVERVIEW OF TRAFFIC SIGNAL CONTROL SYSTEMS

- **Real-Time Monitoring:** Collects traffic information from sensors and detection devices.
- **Input Processing:** Identifies vehicle presence and pedestrian crossing requests.
- **Control Unit:** Uses an embedded microcontroller to process incoming signals.
- **Signal Phase Management:** Coordinates red, yellow, and green lights to prevent conflicting movements.
- **Dynamic Timing:** Adjusts signal durations according to programmed rules and detected traffic conditions.
- **RTOS Coordination:** Schedules control tasks and responds to timer events and interrupts.

TYPE OF OPERATING SYSTEM AND ITS JUSTIFICATION

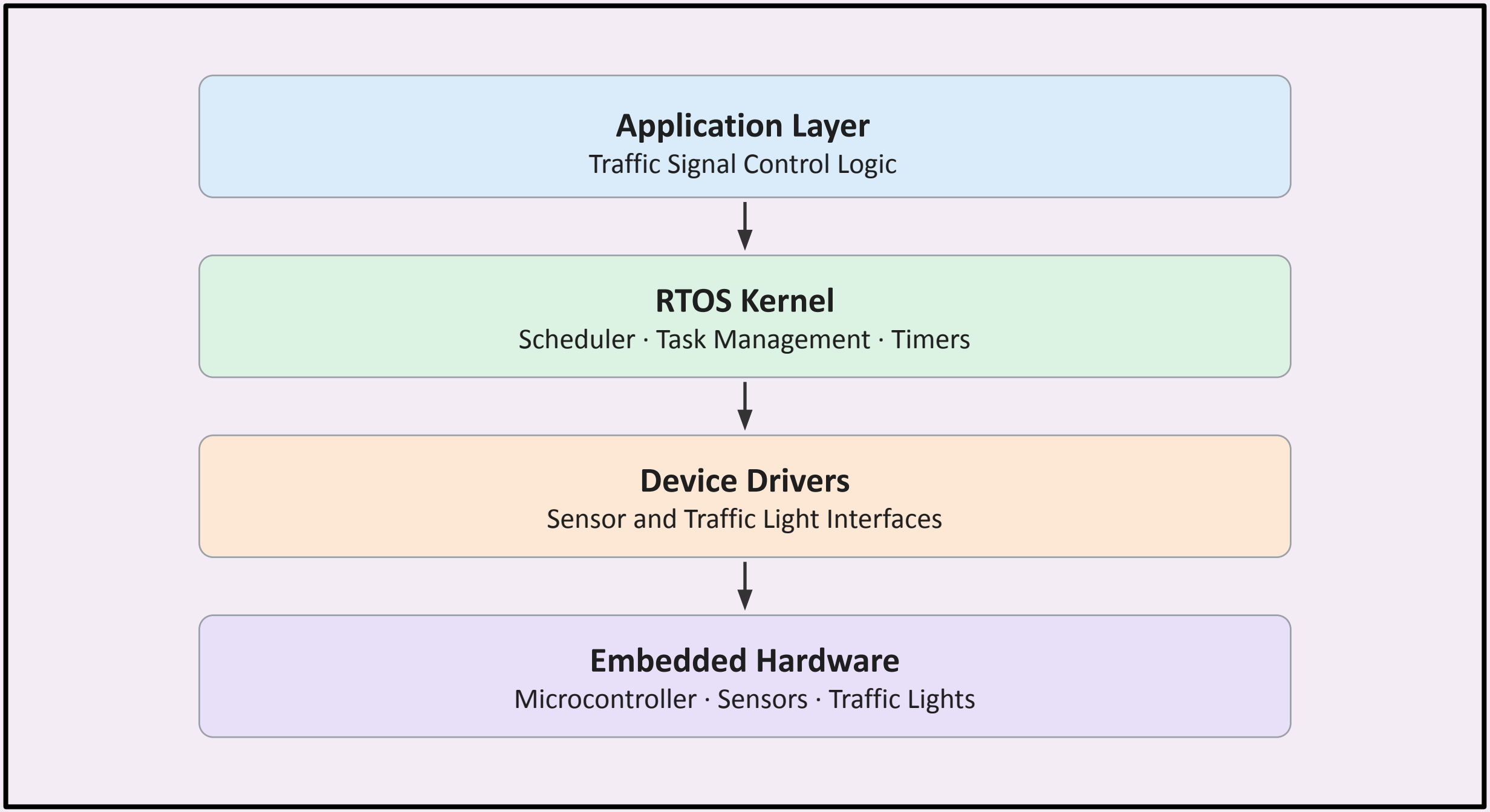
- **Predictable Response:** Executes important tasks within defined time limits.
- **Priority-Based Scheduling:** Gives higher priority to critical traffic control tasks.
- **Timer Management:** Controls the duration of red, yellow, and green signals.
- **Interrupt Handling:** Responds quickly to sensor inputs and timer events.
- **Resource Management:** Manages CPU time, memory, and hardware devices.
- **Reliability:** Supports consistent and coordinated traffic signal operation.

RTOS ARCHITECTURE AND COMPONENTS

- An RTOS manages the hardware and software tasks of a traffic signal control system to ensure timely and reliable operation.

Main Components

1. Application Layer: Executes traffic signal control logic.
2. RTOS Kernel: Manages tasks, scheduling, and system resources.
3. Task Scheduler: Selects which task should execute next.
4. Interrupt Handler: Responds to sensor inputs and timer events.
5. Clock and Timer Services: Manage signal durations and scheduled operations.


Application Layer
Traffic Signal Control Logic
RTOS Kernel
Scheduler · Task Management · Timers
Device Drivers
Sensor and Traffic Light Interfaces
Embedded Hardware
Microcontroller · Sensors · Traffic Lights

PROCESS MANAGEMENT AND CONTROL SYSTEM CALLS

Process Management

- Task Creation: Creates tasks for traffic light control and sensor monitoring.
- Task Scheduling: Executes tasks according to their priority.
- Task Suspension: Temporarily pauses a task.
- Task Resumption: Restarts a suspended task.
- Task Synchronization: Coordinates multiple tasks to avoid conflicts.

API (FreeRTOS)	Function
`xTaskCreate()`	Creates a new task
`vTaskDelete()`	Deletes a task
`vTaskSuspend()`	Suspends a task
`vTaskResume()`	Resumes a task
`xQueueSend()`	Sends data between tasks
`xQueueReceive()`	Receives data between tasks

TIMING SYSTEM CALLS AND REAL-TIME SCHEDULING

Timing System Calls / APIs

- vTaskDelay() – Delays a task for a specified time.
- vTaskDelayUntil() – Executes tasks at regular intervals.
- xTimerCreate() – Creates a software timer.
- xTimerStart() – Starts a timer.
- xTimerStop() – Stops a timer.

Real-Time Scheduling

- Priority-Based Scheduling: Executes high-priority tasks first.
- Periodic Scheduling: Runs control tasks at regular intervals.
- Deadline Management: Helps tasks finish within required time limits.
- Interrupt Handling: Responds quickly to sensor inputs and timer events.
- Task Coordination: Ensures traffic signal operations occur in the correct sequence.

CONCLUSION

- The RTOS architecture coordinates the processor, sensors, traffic lights, timers, and other hardware components.
- Process management and control system calls help execute and coordinate traffic control tasks efficiently.
- Timing system calls ensure that red, yellow, and green signals operate according to their programmed durations.
- Proper task scheduling and timing management improve the reliability and consistency of traffic signal operation.
- Overall, an RTOS supports safer traffic management by providing accurate timing, efficient task coordination, and predictable system responses.

THANK YOU